

x, y in S

1. Set $z = \lambda x + (1-\lambda)y$

2. $x \geq 3$ & $y \geq 3$ state knowns

3. scale: $\lambda \geq 0$, $\lambda x \geq 3\lambda$ since $1-\lambda \geq 0$, $(1-\lambda)y \geq 3(1-\lambda)$

4. add $z = \lambda x + (1-\lambda)y \geq 3\lambda + 3 - 3\lambda = 3$

5. $z \geq 3$ so z is in S and S is convex (both weights need to be nonnegative)

3.

a. $f(x) = 3x^2 + 4x - 5 \rightarrow f''(x) = 6$ Convex

b. $f(x) = 4 - 5x + 3x^2 \rightarrow f'(x) = -5 + 6x \rightarrow f''(x) = 6 \rightarrow$ Convex

c. $f(x) = \exp(x^2) \rightarrow f'(x) = e^{x^2} \rightarrow f''(x) = 2x e^{x^2} \rightarrow$ both convex

d. $f(x) = 7x - 15 \rightarrow f'(x) = 7 \rightarrow f''(x) = 0 \rightarrow$ both (linear function)

(c. $e^{x^2} \rightarrow 2x e^{x^2} \rightarrow 4x^2 e^{x^2}$) for use

2. Express $(2, 2)^T$ as convex combination of $(0, 0)^T$, $(1, 4)^T$ & $(3, 1)^T$

$$(2, 2) = a_1(0, 0) + a_2(1, 4) + a_3(3, 1)$$

First ~~argue~~ $a_2 * 1 + a_3 * 3 = 2$

$$a_2 * 4 + a_3 * 1 = 2$$

$$a_2 * 1 + a_3 * 3 = a_2 * 4 + a_3 * 1$$

$$a_2 + 3a_3 = 4a_2 + a_3$$

$$2a_3 = 3a_2$$

$$a_2 + 2a_3 = 2 \quad 3a_3 = 2$$

$$a_2 = \frac{2}{3} \quad 4a_2 + \frac{2}{3} = 2$$

$$4a_2 = \frac{4}{3}$$

$$a_2 = \frac{1}{3}$$

$$\checkmark a_1 = a_2 + a_3 = 1$$

$$a_1 = 0$$

$$a_3 = \frac{2}{3} \quad \frac{2}{3}$$

$$q. 2. (2, 2) = a_1(0, 0) + a_2(1, 4) + a_3(3, 1)$$

$$a_2 + 3a_3 = 2 \rightarrow a_3 = \frac{2}{3} - \frac{1}{3}a_2$$

$$4a_2 + a_3 = 2 \quad \leftarrow \text{check } 0 \leq 20$$

$$4a_2 + \frac{2}{3} - \frac{1}{3}a_2 = 2$$

$$3\frac{2}{3}a_2 = 1\frac{1}{3}$$

$$11/3 a_2 = 4/3 \quad \text{a line}$$

$$11a_2 = 4$$

$$a_2 = \frac{4}{11}$$

$$\frac{4}{11} + 3a_3 = 2$$

$$3a_3 = \frac{22}{11} - \frac{4}{11} = \frac{18}{11}$$

$$a_3 = \frac{6}{11}$$

$$a_1 = \frac{1}{11}$$

$$= 1 - \frac{4}{11} - \frac{6}{11} =$$

$$6. x_1 + x_2 = 1, x_1 \geq 0, x_2 \geq 0$$

$$a. (0, 1)$$

$$a1. 0 + 1 = 1 \quad 0 = 0 \quad 1 > 0 \quad \text{first 2 active}$$

$$p_1 + p_2 = 0$$

$$p_1 \geq 0$$

$$p_2 = -p_1$$

$$p_1 = + \leftarrow \text{free variable}$$

$$p_2 = -t \quad t \geq 0$$

$$p = (+, -) = t(1, -1) \quad \text{with } t \geq 0$$

$$b. (1, 0)$$

$$1 + 0 = 1 \quad 1 \geq 0$$

$$0 = 0$$

$$1 \& 3rd \text{ active}$$

$$p_1 + p_2 = 0$$

$$p_2 \geq 0$$

$$p_1 = -p_2$$

$$p_2 = +$$

$$p_1 = -t$$

$$t \geq 0$$

$$p(-t, +) = t(-1, 1) \quad \text{with } t \geq 0$$

6. $(0.5, 0.5)$ $(1, 1) \leq 0 + (1, 1) \leq 0 + (0, 0), 0 = (5, 5)$
 $50 \frac{1}{2} - \frac{5}{2} = 80 \leftarrow S = 20x_1 + 10x_2$
 $0.5 + 0.5 = 1 \quad 0.5 > 0 \quad 0.5 > 0$ Only 0/1 active

$p_1 + p_2 = 0$
 $p_1 = -p_2 \quad p_1 = +$ no inequality restricting it
 $p_2 = -+$ therefore
 $p(-+, +)$ with $(+ \in \mathbb{R})$

4. $f(x) = -x_1 - x_2$ subject to $x_1 + x_2 \leq 2$ and $x_1, x_2 \geq 0$.
 a. feasible directions at $(0, 0), (0, 1), (1, 0) \& (0, 2)$
 aa. $(0, 0)$

~~$0 + 0 < 2$~~ $0 + 0 < 2 \quad x_1 = 0 \quad x_2 = 0$ 2 & 3 active
 $p_1 \geq 0 \quad p_2 \geq 0$
 $\therefore p(0, 0)$ $\leftarrow$ you can move anywhere

ab. $(0, 1)$

$0 + 1 < 2 \quad x_1 = 0 \quad x_2 = 0$
 $p_1 \geq 0 \quad p_2 = +$
 ~~$p(0, 1)$ with $(+ \in \mathbb{R})$~~
 $p(0, +)$ with $(+ \in \mathbb{R})$

ac. $(1, 1)$

active slack slack
 $p_1 = -p_2 \quad p_1 + p_2 \leq 0$
 $p = (-+, +)$ with $(+ \in \mathbb{R})$

ad. $(0, 2)$

active active slack
 $p_1 + p_2 \leq 0 \quad p_1 \geq 0$

Local minimizers

$(-1, -1) \quad (1, 0)$

$-p_1 - p_2 < 0$

$0 = 2q + 1q$

$1q = -2q$

$3q = 0$

$q = 0$

$(0, 1)$

$0 \leq 1 - 1 = 0$

$0 = 1q + 1q$

$1q = -1q$

$2q = 0$

$q = 0$

$(1, 1)$

$(1, 1)$